Highlights

Popularity Bias and Cold-Start in Patent Technology-Transfer Demand Prediction: A Diagnostic Evaluation

- Under a temporal, same-IPC hard-negative protocol, no learned model beats a popularity baseline on KIPRIS patent-transfer prediction.
- The failure is traced to popularity bias (score–popularity ρ up to 1.0) and a 91.7% patent cold-start regime.
- A strict tie-break inflates degenerate cold-start models to near-perfect scores; average-rank tie-breaking is required.
- A random split reverses the verdict (SVD becomes best at 0.442), showing that the protocol, not the model, determines the outcome.
- Standard popularity-bias mitigations (IPS, logQ, debiased sampling) give only partial, unstable gains.

Popularity Bias and Cold-Start in Patent Technology-Transfer Demand Prediction: A Diagnostic Evaluation

Abstract

Predicting which company will acquire the rights to a given patent — patent technology-transfer demand prediction — can be framed as link prediction on a heterogeneous graph coupling a patent–cites–patent network with patent–transfer–company edges. We evaluate this task on the Korean KIPRIS dataset (370,666 patents; 122,519 companies) under a realistic protocol: transfer edges are split temporally by registration date, candidates are same-IPC hard negatives, and ties are resolved by average rank. Under this protocol every learned model we test — heterogeneous GNNs (GraphSAGE, GAT), collaborative filtering (LightGCN, NGCF), matrix factorization (SVD), structural heuristics (Common Neighbors, Adamic–Adar), and a text MLP — falls below a simple most-popular baseline (MostPop NDCG@10 = 0.197; the strongest learned model, a text MLP, reaches only 0.150), with AUC near chance (0.42–0.60). Two causes account for this: popularity bias (98.8% of GraphSAGE and 57.8% of GAT errors are losses to a more-popular hard negative, and learned-model scores track popularity up to $\rho = 0.96$) and patent cold-start (91.7% of test patents unseen; SVD NDCG@10 = 0.000 on new patents). We document an evaluation artifact whereby a strict greater-than tie-break inflates degenerate cold-start models toward high scores, which average-rank tie-breaking removes. A controlled random-split contrast shows the protocol determines the outcome: a random split lowers cold-start to 2.4% and makes SVD the single best method (NDCG@10 = 0.442, AUC = 0.830), the reversal of the temporal verdict. We contribute the evaluation protocol, a suite of bias and cold-start diagnostics, and a symmetric grid of mitigation baselines that yield only partial, unstable gains.

Keywords: patent technology transfer, link prediction, graph neural networks, popularity bias, cold-start, evaluation methodology

1. Introduction

Patent technology transfer — the licensing, assignment, or sale of patent rights from one party to another — is a central mechanism by which research output is converted into commercial use. For technology-holding institutions and the intermediaries that support commercialization, a recurring operational question is: *given a particular patent, which companies are plausible recipients of its rights?* Answering it well would shorten the search for licensees and help allocate scarce brokerage effort [1, 2]. In the Korean setting, where public technology-transfer programs and the KIPRIS registry make large-scale transaction records observable, the question is both economically meaningful and empirically tractable [3, 4]. We refer to this task as *technology-transfer demand prediction*, with the explicit caveat that the available supervision is realized transfers rather than latent demand.

The task has a natural relational structure. Patents cite other patents and the scientific literature — reference signals rich enough to have motivated dedicated large-scale extraction pipelines [5] — patents are transferred to companies, and companies accumulate portfolios. This induces a heterogeneous graph with *patent* and *company* nodes and two edge types (patent–cites–patent and patent–transfer–company), in which demand prediction becomes link prediction. Cast this way, the problem invites graph representation learning and collaborative filtering: GNNs such as GraphSAGE [6] and graph attention networks [7, 8], CF models such as NGCF [9] and LightGCN [10], and matrix factorization [11], with patent text encoded by sentence embeddings [12] providing node features. The expectation that such methods outperform simple heuristics is the default working hypothesis in much of the literature and motivates a growing line of patent-recommendation systems [14, 15, 16, 17]. Most directly, a recent GAT–NGCF firm–patent recommender on Korean technology-transfer data reports Recall@5 $= 0.998$ and NDCG@5 $= 0.997$ [18]. Such near-perfect figures rest on random-split, sampled-negative evaluation; the gap we document is that they do not survive a leakage-free, temporal, same-IPC hard-negative re-evaluation, which is the research gap this paper addresses.

That expectation is sensitive to how performance is measured. *Random edge splits* break temporal order and leak future transfers [19]; *uniform random negatives* make ranking easy and inflate AUC relative to a deployment setting where all candidates are topically relevant [20]; and headline AUC can mask the popularity bias known to dominate recommendation under skewed exposure [21, 22, 23, 24] — a bias whose effect on accuracy and fairness is itself a function of dataset characteristics [25], and whose corrective methods transfer only partially to practice [26]. A realistic protocol must therefore combine temporal ordering, same-class hard negatives, explicit cold-start accounting, and care over tie-breaking.

In this paper we evaluate the task under such a protocol and report a primarily diagnostic finding: under a temporal 70/15/15 split with same-IPC hard negatives ($n_{\text{neg}} = 100$) and an average-rank tie-break, *every learned model we test falls below simple popularity baselines.* The dataset comprises 370,666 patents, 122,519 companies, and ≈910,000 citation edges, yielding ≈220,284 temporally held-out test queries. A most-popular ranker attains NDCG@10 = 0.197; an IPC-conditional skyline reaches 0.199 (the top NDCG@10, though at a below-chance global AUC of 0.416; see §5.1); and NGCF — which our diagnostics show to be a popularity ranker ($\rho = 0.96$) — reaches 0.196. The strongest learned model with above-chance discrimination is a text MLP at 0.150; among graph models, GraphSAGE+logQ reaches 0.125 and GAT+logQ 0.107; all lie well below the popularity baselines, and the AUC of every model is near or below chance (0.42–0.60). The pattern holds across GNNs, CF, MF, structural heuristics, and a text MLP. It mirrors a broader concern that learned recommenders often fail to beat well-tuned baselines [27] and that easy negatives inflate GNN link-prediction [28].

We trace the result to two causes and quantify each. First, *popularity bias*: 98.8% of GraphSAGE and 57.8% of GAT errors are losses to a more-popular same-IPC hard negative; the hard-negative inversion rate (the share of queries in which a more-popular candidate outranks the positive) is 56% for MostPop and NGCF and 28% for GAT; and score–popularity Spearman correlation is 1.0 for MostPop and 0.96 for NGCF. A GAT attention analysis (Mann–Whitney U over hub vs. non-hub edges, $p = 1.0$) shows no down-weighting of popular hubs. Second, *extreme patent cold-start*: 91.7% of test patents are unseen in training, and SVD collapses to NDCG@10 = 0.000 on new patents. We then test whether popularity-bias mitigations recover learned models above the baselines; across more than 20 configurations the gains are partial and unstable (§6).

A further contribution is methodological. Under a strict greater-than tie-break, cold-start models whose scores are degenerate place the positive first among tied candidates and attain inflated scores (§5.5); an average-rank tie-break removes this artifact. These observations motivate three research questions addressed in the paper:

- **RQ1.** Does a realistic protocol (temporal split + same-IPC hard negatives) change measured performance relative to a random split? (§5.7: yes.)
- **RQ2.** What accounts for the performance — popularity bias, cold-start, or both — and how is each quantified? (§5.2–§5.4.)
- **RQ3.** Can popularity-bias mitigations recover learned models above the popularity baselines? (§6: not reliably.)

The contributions are: (i) a leakage-free evaluation protocol for the task on KIPRIS, combining a temporal split, same-IPC hard negatives, explicit cold-start accounting, and an average-rank tie-break; (ii) a diagnosis that GNNs, CF, MF, structural heuristics, and a text MLP all fall below popularity baselines, with AUC near chance, together with the quantification of popularity bias and cold-start that explains why; (iii) a reusable suite of bias-diagnostic tools that localize the failure; (iv) a mitigation grid showing only partial, unstable gains; and (v) identification of a strict-tie-break artifact and its average-rank correction, corroborated by an unsampled full-pool metric and a temporal-versus-random contrast. The aim is not a winning model but an evaluation protocol and diagnosis to discipline future work. Figure 1 gives an overview.

2. Related Work

Our study sits at the intersection of four lines of work: patent analytics and technology-transfer prediction; GNNs for link prediction and recommendation; popularity bias and debiasing; and evaluation methodology for ranking under temporal and sampled-metric constraints. We also use structural link-prediction heuristics as non-learned baselines. The recurring theme is that this paper's contribution is diagnostic: we do not propose a model that outperforms prior art, but an evaluation protocol and diagnostic tools that explain *why*, on this task and dataset, learned models do not separate from popularity baselines.

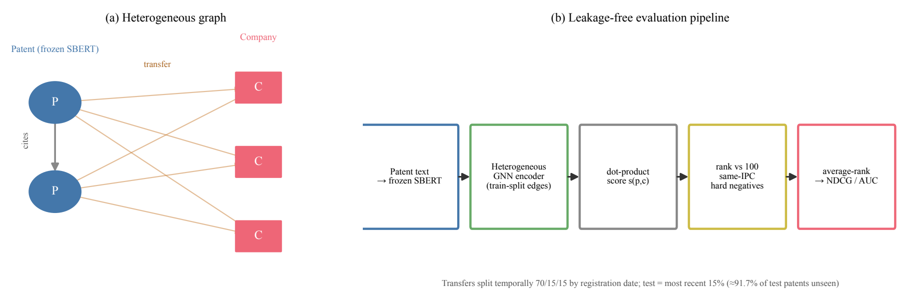

Figure 1: Overview of the task and the central finding. We frame patent technology-transfer demand prediction as link prediction on a heterogeneous patent–citation–transfer graph and evaluate it under a leakage-free protocol (temporal split; encoding on training-split edges only; ranking the true acquirer against 100 same-IPC hard negatives; average-rank tie-break). Under this protocol every learned model falls below a most-popular baseline and discriminates near chance, which we trace to popularity bias and an extreme (91.7%) patent cold-start regime; a conventional random split would reverse this verdict.

2.1. Patent Analytics and Technology-Transfer Prediction

A substantial body of work uses patent metadata, classification codes, and citation networks — including recent work on extracting and matching patent in-text references at scale [5] — to characterize technological trajectories and forecast emerging technologies [14, 37]. Within this literature, technology transfer has been examined as a prediction target [1, 3], with approaches ranging from feature-engineered and BERT classifiers [1] to collaborative filtering over IPC/portfolio structure [4, 2] and graph formulations [18, 16, 17]. Several graph recommenders report near-perfect retrieval (e.g., Recall@5 = 0.998 [18]) under random-split, sampled-negative evaluation — precisely the regime our protocol revisits. Three properties distinguish our setting: the supervision is a *realized* transfer; the prediction is prospective; and the patent side is severely cold-start (91.7% of test patents are new). Table 1 contrasts the evaluation safeguards our protocol applies with those prevailing in prior graph-based patent recommenders.

2.2. GNNs for Link Prediction and Recommendation

GraphSAGE [6] introduced inductive aggregation; GAT [7] learned attention weights, and GATv2 [8] corrected a static-attention limitation. In recommendation, LightGCN [10] simplified propagation, whereas NGCF [9] retains

Table 1: Evaluation safeguards in the protocol of this paper versus those typically applied in prior graph-based patent-recommendation studies (e.g., [18, 16, 17]). "Typical" denotes the prevailing practice, not any single paper. Each safeguard counters a specific source of optimism (future leakage, easy negatives, tie degeneracy, an absent skyline, hidden cold-start, or a single arbitrary cutoff).

Evaluation safeguard	Typical prior eval	This work
Temporal (forward-in-time) split	usually random	yes
Hard negatives from the same IPC class	uniform / in-batch	yes
Average-rank tie-break	unspecified	yes
Popularity skyline as the bar to beat	weak / absent	yes
Cold-start fraction reported and decomposed	rarely	yes
AUC reported beside top-K	top-K only	yes
Unsampled full-IPC-pool metric	no	yes
Rolling-origin (multiple temporal cutoffs)	no	yes
Multi-seed, paired tests, multiplicity control	rare	yes

feature transformation; DropEdge [33] is a structural regularizer. Heterogeneous/relational variants such as R-GCN [34] and the Heterogeneous Graph Transformer [35] extend message passing to typed graphs; SEAL [36] frames link prediction as subgraph classification. We adopt these architectures essentially as published and ask whether, under a leakage-free temporal split with same-IPC hard negatives, they separate from non-learned baselines. They do not; §5.2 examines why.

2.3. Popularity Bias and Debiasing

A well-documented failure mode of implicit-feedback recommenders is popularity bias: models score frequently interacted items highly regardless of relevance, and standard metrics reward this [21, 22]. IPS methods reweight by inverse exposure probability [30, 31]; sampling-bias-corrected training (logQ) adjusts sampled-softmax logits by log-frequency [32]; and the BPR framework [11] underlies much of this work. Within the present venue, a line of work has examined this bias directly: Boratto et al. [24] connect user- and item-side perspectives on popularity debiasing; Deldjoo et al. [25] show the accuracy/fairness consequences are governed by dataset characteristics; and Boratto et al. [26] report that consumer-fairness mitigations transfer only partially to practice — a partial-and-unstable pattern matching our own results. Popularity bias is the central finding of our diagnosis, made explicit rather than assumed (§5.2).

2.4. Evaluation Methodology

Offline recommender evaluation is fragile: many neural recommenders fail to beat tuned baselines [27, 38]; sampled top-K metrics can be inconsistent with full ranking [20, 39, 40]; random splits leak future information relative to temporal splits [19, 41]; and easy negatives inflate GNN link prediction [28]. Hidasi and Czapp [29] catalogue widespread offline-evaluation flaws, including degeneracies of the kind our tie-break artifact exemplifies. Our protocol operationalizes these recommendations and adds an average-rank tie-break necessary to avoid inflating no-information cold-start models.

2.5. Structural Heuristics and Cold-Start

Common Neighbors and Adamic–Adar are strong, parameter-free baselines for homogeneous link prediction. Cold-start, where items carry no interaction history, is addressed by content- and meta-learning methods [42]; our setting is an extreme instance (91.7% new patents). We include Common Neighbors and Adamic–Adar over the citation graph; both fall below the popularity skylines (NDCG@10 = 0.065–0.066), confirming that citation-structure overlap alone does not recover the transfer signal.

3. Dataset, Heterogeneous Graph, and Models

3.1. Data

The KIPRIS patent data is obtained from the Korea Intellectual Property Rights Information Service API. We use three tables: `patents` (application number, IPC code, title, abstract), `transfers` (application number, acquiring company, registration date), and `citings` (citing/cited application numbers). Application numbers are cleaned to digits only before joining. The corpus contains 370,666 patents, 122,519 companies, and ≈910,000 citation edges. Patent text (title + abstract) is encoded once with a frozen multilingual Sentence-BERT model (`paraphrase-multilingual-MiniLM-L12-v2`) [12, 13] into 384-dimensional features, row-aligned to the patent table.

Table 2: Notation.

Symbol	Meaning
$G = (V, E)$, $V = P \cup C$	heterogeneous graph; patent / company nodes
$E_{\text{cite}}, E_{\text{tr}}$	citation edges; realized transfer edges
$x_p \in \mathbb{R}^{384}$, $x_c \in \mathbb{R}^{64}$	frozen SBERT patent / company feature
$h_p, h_c \in \mathbb{R}^{16}$	encoder node representations
$s(p, c) = h_p^\top h_c$	dot-product link score
$C^-(p)$, n_{neg}	same-IPC hard negatives; candidate size (100)
$\text{rank}(c^+ \mid p)$	average rank of the positive among candidates

3.2. Heterogeneous Graph and Problem Formulation

We model the task as link prediction on a heterogeneous graph $G = (V, E)$ with $V = P \cup C$ (P patents, C companies) and two undirected-typed edge sets: citations $E_{\text{cite}} \subseteq P \times P$ and realized transfers $E_{\text{tr}} \subseteq P \times C$. Each patent carries its frozen 384-d SBERT feature x_p. For the GNN encoders each company is assigned a *fixed-random* feature $x_c \in \mathbb{R}^{64}$ (drawn once as $\mathcal{N}(0, 0.1^2)$, never updated), with the only learned company-side parameters being a shared linear projection; only LightGCN/NGCF use a learnable per-company identity embedding ($d = 64$). We ablate this random-feature choice with a leakage-free content company representation in §5.8. An encoder f_θ maps nodes to $h_p, h_c \in \mathbb{R}^{16}$ by message passing over E_{cite} and the *training-split* transfers only, and a parameter-free decoder scores a pair by the dot product $s(p, c) = h_p^\top h_c$; this is the same score optimized by the training loss (§3.3) and re-ranked by the IPS / logQ mitigations (§6). For a held-out test transfer (p, c^+) we form $\text{Cand}(p) = \{c^+\} \cup C^-(p)$, where $C^-(p)$ are $n_{\text{neg}} = 100$ same-IPC hard negatives — companies that received a same-IPC patent in training but not p — and report the average rank of c^+. Table 2 summarizes the notation. A construct-validity caveat underlies the study: the supervision is *realized* transfers, so a model that perfectly predicted them would recover the historical (and biased) matching process, not latent demand.

3.3. Models Evaluated

We evaluate nine base models: three non-learning skylines (MostPop, IPC-conditional MostPop-IPC, and Recency, which ranks candidate companies by the recency of their most recent training transfer); two structural heuristics (Common Neighbors, Adamic–Adar); SVD (k=64); a text-only

MLP; and two CF models (LightGCN, NGCF); together with two heterogeneous GNNs (GraphSAGE, GAT), and a *symmetric mitigation grid* applying each of {Debias, logQ, DropEdge, time encoding, IPS} and two stacked combinations (Debias+IPS, Time+IPS) to *both* GNN backbones. GAT uses a single-head, scatter-based attention layer that can return attention weights for diagnostics. Training minimizes a binary cross-entropy loss on the dot-product scores $s(p, c) = h_p^\top h_c$ of observed transfers (label 1) and sampled negatives (label 0), i.e. $\mathcal{L} = -\sum_{(p,c^+)} \log \sigma(s(p, c^+)) - \sum_{(p,c^-)} \log(1 - \sigma(s(p, c^-)))$, with early stopping on validation NDCG@10 (patience 5).

4. Evaluation Protocol

Temporal split.. Transfers are sorted by registration date and split 70/15/15 into train/validation/test. The encoder message-passes only on training transfers; popularity, recency, and time features are computed from the training split alone, removing the target leakage a random split would introduce.

Same-IPC hard negatives.. For each test query $(p, c^+, \text{ipc4})$, the candidate set is the positive plus $n_{\text{neg}} = 100$ companies drawn (seed-fixed) from those that received a same-IPC patent in training but not p (average padding rate 1.07% when fewer than 100 same-IPC companies exist; padding fills the candidate set with uniformly drawn non-positive companies so every query has exactly n_{neg} negatives).

Metrics and tie-break.. We report Hits@K and NDCG@K for $K \in \{1, 3, 5, 10\}$, MRR, and a tie-aware rank-AUC (ties counted as 0.5). Ties are resolved by *average rank*: a candidate that ties the positive contributes 0.5 to its rank. This is necessary, because under a strict greater-than rule a no-information model whose candidates all tie places the positive first and scores spuriously high (§5.5).

Statistics.. Every configuration is run over ten seeds. We test a pre-registered family of pairwise comparisons with the Wilcoxon signed-rank test (primary) and paired t-test (secondary), correcting jointly with Holm–Bonferroni, and report percentile bootstrap 95% confidence intervals over the ≈220k test queries.

Reproducibility.. Seeds (torch, numpy) are fixed per run. The full ten-seed run completes in ≈70 minutes on a single NVIDIA Tesla T4 and, comparably, on Apple-Silicon CPU; LightGCN/NGCF use sparse operations unsupported by Apple MPS, so the intended device is CUDA or CPU. We release the preprocessing and evaluation code and a script to regenerate the SBERT embeddings.

5. Results and Diagnosis

5.1. Main Results

Table 3 reports the main ten-seed results. *No learned model surpasses a simple popularity baseline.* The top three configurations are non-learned or popularity-equivalent: MostPop-IPC (0.199), MostPop (0.197), and NGCF (0.196), which our diagnostics (§5.2) show to be a popularity ranker. The strongest learned model with above-chance discrimination is the text MLP (0.150); the best graph model is GraphSAGE+logQ (0.125), and GAT+logQ reaches 0.107 — all below the popularity baselines. (The IPS-penalized variants reach 0.157–0.158, but only by driving AUC below chance; §6.) SVD (0.037), the structural heuristics (0.065–0.066), LightGCN (0.058), and both bare GNNs (GraphSAGE 0.084, GAT 0.074) are far below. The AUC of every model lies between 0.42 and 0.60 — near or below chance — confirming that none globally separates true transfers from same-IPC alternatives. Figure 2 plots all models.

5.2. Popularity Bias (RQ2)

Three diagnostics localize the failure to popularity bias (Table 4, Figure 3). (i) The Spearman ρ between a model's candidate scores and the candidates' training popularity is 1.0 for MostPop and 0.96 for NGCF — NGCF ranks almost exactly like a popularity counter, which is why its NDCG@10 (0.196) matches MostPop's. GAT has $\rho = -0.05$. (ii) A more-popular negative is ranked above the true positive in 56.4% of MostPop and 56.3% of NGCF queries, versus 28.4% for GAT. (iii) A Mann–Whitney U test comparing GAT's attention on edges to popular (top-5%) hub companies versus non-hub companies yields $p = 1.0$: GAT does *not* down-weight popular hubs. Stratifying GAT's accuracy by company popularity shows it does worst on the popular head (NDCG@10 = 0.016, torso 0.082) and best on the unpopular tail (0.370) — the opposite of a popularity-exploiting model, yet its absolute

Table 3: Main results, temporal split, 10 seeds (mean). The three top NDCG@10 rows are non-learned or popularity-equivalent. MostPop-IPC tops NDCG@10 (0.199) yet has a below-chance AUC (0.416): ranking by same-IPC-conditional popularity places the positive well within the top-K of its own IPC class, but inverts the global pairwise ordering of positives against out-of-class candidates, so the two metrics diverge. Full 25-row table (with standard deviations and all mitigation variants) in the supplementary results file.

Model	Hits@1	Hits@10	MRR	NDCG@10	AUC
MostPop	0.110	0.302	0.182	0.197	0.583
MostPop-IPC	0.122	0.289	0.184	0.199	0.416
Recency	0.029	0.282	0.128	0.149	0.602
Common Neighbors	0.056	0.070	0.082	0.065	0.534
Adamic–Adar	0.057	0.070	0.082	0.066	0.534
SVD	0.030	0.044	0.053	0.037	0.520
MLP (text)	0.082	0.234	0.142	0.150	0.574
LightGCN	0.019	0.115	0.063	0.058	0.509
NGCF	0.110	0.301	0.182	0.196	0.584
GraphSAGE	0.042	0.141	0.087	0.084	0.473
GAT	0.052	0.102	0.085	0.074	0.519
GAT + logQ	0.094	0.127	0.121	0.107	0.548
GraphSAGE + logQ	0.099	0.162	0.133	0.125	0.540
GAT + IPS	0.143	0.181	0.166	0.157	0.417
GraphSAGE + Debias+IPS	0.145	0.179	0.167	0.158	0.417

accuracy stays far below MostPop because the head holds most positives, so the head-stratum score dominates the aggregate (Figure 4).

5.3. Patent Cold-Start (RQ2)

Under the temporal split, 91.7% of test patents are unseen in training. Decomposing accuracy by patent novelty matters most for SVD: on the new-patent subset its NDCG@10 is 0.000 (its latent factor for an unseen patent is zero), whereas on the seen-patent subset it reaches 0.439. Because new patents dominate, SVD's overall score collapses to 0.037. The popularity baselines degrade gracefully (MostPop new-patent 0.212 vs. seen 0.324) because a popular company remains a good guess regardless of patent novelty; GAT is weak on both (0.022 / 0.021). Figure 5 visualizes this.

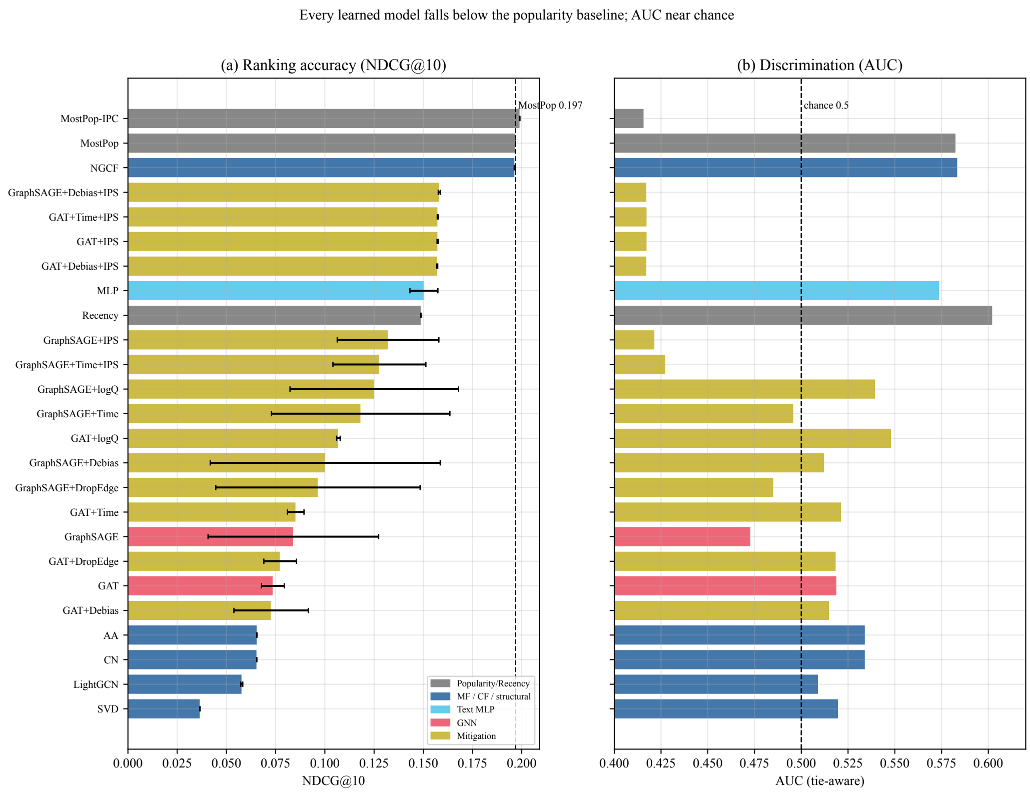

Figure 2: Main results (temporal split). (a) NDCG@10 of all models, sorted; the dashed line is the MostPop skyline (0.197). (b) Tie-aware AUC; the dashed line is chance (0.5). Every learned model falls below the popularity baseline and discriminates near chance.

5.4. Error-Source Decomposition (RQ2)

Attributing each failure to a cause (priority: popularity > rare/new > residual) shows the backbones fail differently (Figure 6). For GraphSAGE, 98.8% of failures are losses to a more-popular hard negative (0.1% rare/new, 1.1% residual): it fails almost entirely through popularity. For GAT, the shares are 57.8% / 8.3% / 33.9% — popularity still dominates, but a third of GAT's errors are a semantic residual, consistent with its near-zero popularity correlation. The two backbones fail through different mechanisms.

5.5. The Tie-Break Artifact

When many candidates receive identical scores — the norm for cold-start models whose representations are degenerate — the tie-break rule determines

Table 4: Popularity-bias diagnostics (temporal).

Model	Spearman ρ (score, popularity)	Hard-neg inversion rate
MostPop	1.00	56.4%
NGCF	0.96	56.3%
Recency	0.69	50.0%
MLP	0.19	19.0%
GraphSAGE	0.07	51.5%
SVD	0.02	2.4%
GAT	−0.05	28.4%

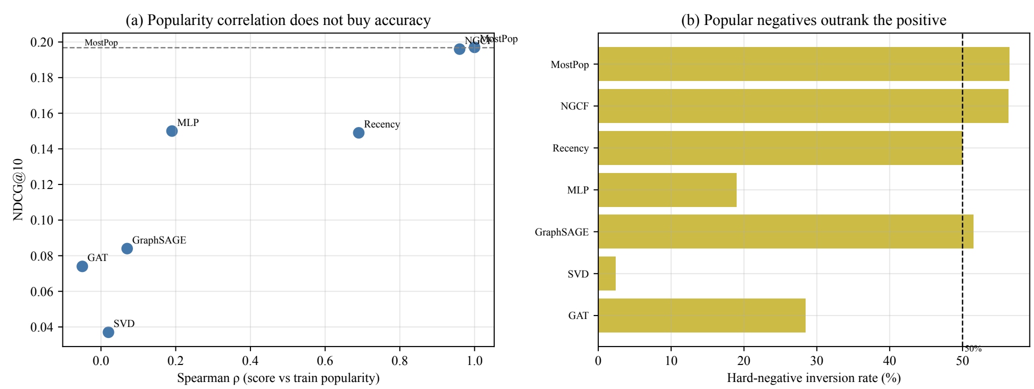

Figure 3: Popularity bias. (a) Score–popularity Spearman ρ vs. NDCG@10: high correlation (MostPop, NGCF) coincides with high accuracy, but no learned model with low correlation recovers it. (b) Hard-negative inversion rate: a more-popular negative outranks the positive in 56% of MostPop/NGCF queries.

the metric. Under a strict greater-than rule the positive is placed ahead of every tied candidate *by construction*: because 91.7% of test patents are cold and a fully-degenerate model assigns every candidate the same score, such a model is ranked first on essentially every cold query and appears near-perfect. In an earlier strict-rule evaluation this inflated SVD (whose latent factor for an unseen patent is zero, leaving all candidates tied) to NDCG@10 $\approx$ 0.95, and a fully-tied GraphSAGE+logQ to 1.00 by construction. Replacing the strict rule with an *average-rank* tie-break (a tie contributes 0.5 to the positive's rank, so a fully-tied candidate set yields rank $(n_{\text{neg}}+1)/2$, i.e. chance) removes the artifact and returns these models to their true ten-seed levels: SVD $=$ 0.037 and GraphSAGE+logQ $=$ 0.125 (and the partially-degenerate

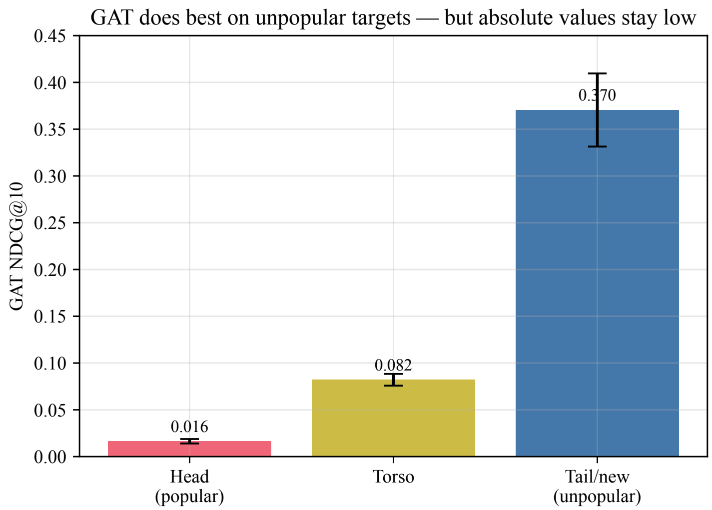

Figure 4: GAT NDCG@10 stratified by company popularity. GAT does best on the unpopular tail and worst on the popular head, but absolute values stay far below MostPop.

GAT from ≈0.45 to 0.074). Figure 7 contrasts the rules. This is a generic pitfall for sampled-ranking evaluation of cold-start link prediction.

5.6. Robustness to the Sampled Metric

Two checks (both seed-0, since they are seed-invariant re-scorings of trained models) confirm the verdict is not an artifact of the 100-negative sample. First, a candidate-set-size sweep ($n_{\text{neg}} \in \{50, 100, 200\}$) preserves the ordering: MostPop NDCG@10 = 0.255 / 0.196 / 0.152 dominates GAT (0.104 / 0.082 / 0.069) at every size. Second, an *unsampled full-IPC-pool* ranking — scoring each positive against the entire eligible same-IPC pool (mean 900 candidates, capped at 1000; 78% of queries have pools exceeding 1000) — preserves it too: MostPop = 0.078, GAT = 0.058, SVD = 0.028, GraphSAGE = 0.008. Even against ≈9× more candidates and with no sampling, no learned model beats MostPop [20].

5.7. The Protocol Is Decisive: Temporal vs. Random (RQ1)

To quantify how much the protocol matters, we re-ran every model under a conventional random 70/15/15 split (Table 5, Figure 8). A random split lowers the patent cold-start rate from 91.7% to 2.4%, because patents are scattered across train and test rather than held out by time. Under it,

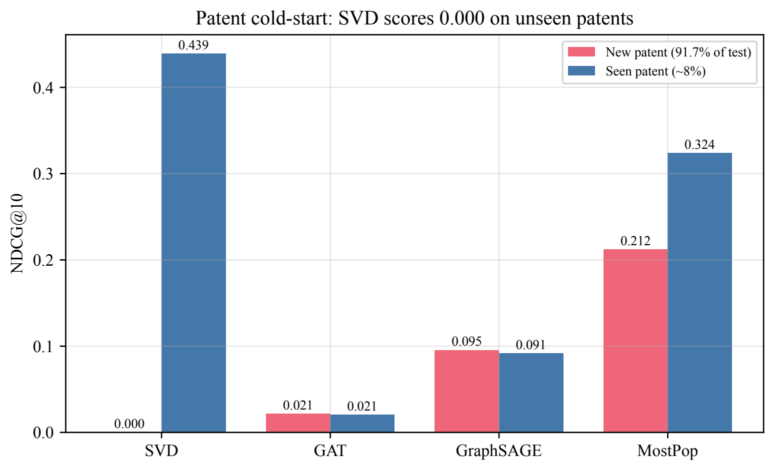

Figure 5: Patent cold-start. NDCG@10 on the new-patent (91.7% of test) vs. seen-patent subsets. SVD scores 0.000 on new patents; the popularity baselines degrade gracefully.

Table 5: RQ1 contrast: NDCG@10 (AUC) under temporal vs. random splits.

Model	Temporal	Random
MostPop	0.197 (0.583)	0.219 (0.667)
MostPop-IPC	0.199 (0.416)	0.290 (0.604)
SVD	0.037 (0.520)	**0.442 (0.830)**
NGCF	0.196 (0.584)	0.216 (0.663)
GAT	0.074 (0.519)	0.165 (0.650)
Patent cold-start rate	91.7%	2.4%

matrix factorization becomes the single strongest method: SVD rises from NDCG@10 = 0.037 (AUC 0.52) to 0.442 (AUC 0.830), comfortably exceeding MostPop (0.219), reversing the temporal verdict. AUC inflates across the board (MostPop 0.583 → 0.667; GAT 0.519 → 0.650). The mechanism is straightforward: with only 2.4% new patents, SVD's latent factors (useless on truly-unseen patents — its random-split new-patent NDCG@10 is still 0.0002) almost always apply. A random-split evaluation would therefore have reported a clear learned-model advantage; the realistic protocol, by exposing the cold-start regime the random split conceals, reverses this outcome. This is the empirical core of RQ1 and the reason near-perfect numbers in prior random-split patent recommenders [18] do not contradict our diagnosis.

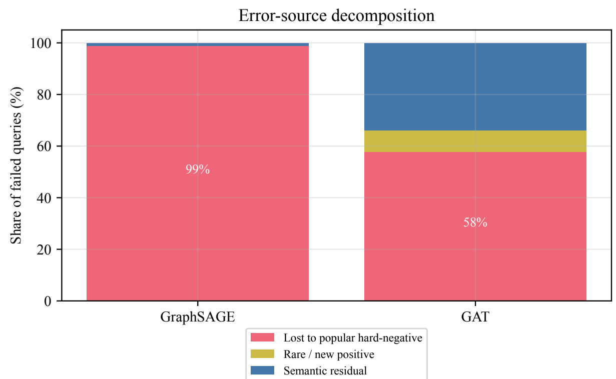

Figure 6: Error-source decomposition. Share of failed queries (rank > 1) attributable to each cause. GraphSAGE fails almost entirely through popularity; GAT spreads errors across popularity and a semantic residual.

5.8. Content company-feature ablation

A natural objection is that the GNNs underperform only because their company nodes carry no content. We therefore replace the fixed-random company feature with the mean frozen-SBERT embedding of each company's training-received patents — a leakage-free content representation (the 23.7% of companies with no training transfer keep a random fallback) — and rerun the temporal protocol. Content company features do not lift any learned model above the popularity baselines (Table 6): they leave the bare GNNs essentially unchanged, and slightly lower (GraphSAGE $0.084 \rightarrow 0.062$, GAT $0.074 \rightarrow 0.051$), and they sharply degrade the text MLP ($0.150 \rightarrow 0.032$, AUC $0.57 \rightarrow 0.40$), while the popularity skylines, SVD, and the CF models are unaffected by construction (their scores do not use this feature). Uninformative random company features are therefore not the explanation for the diagnosis: even a leakage-free content representation of the company side does not recover the transfer signal.

5.9. Rolling-origin robustness

To check that the diagnosis is not an artifact of the single 70/15/15 cutoff, we repeat the evaluation under a rolling-origin (forward-chaining) protocol with three expanding-window folds, each training on all transfers before a 5% validation slice and testing on a successive 10% time slice (three seeds per fold). The verdict is stable across all three folds (Table 7): the popularity skylines and the popularity-equivalent NGCF lead at every cutoff (MostPop

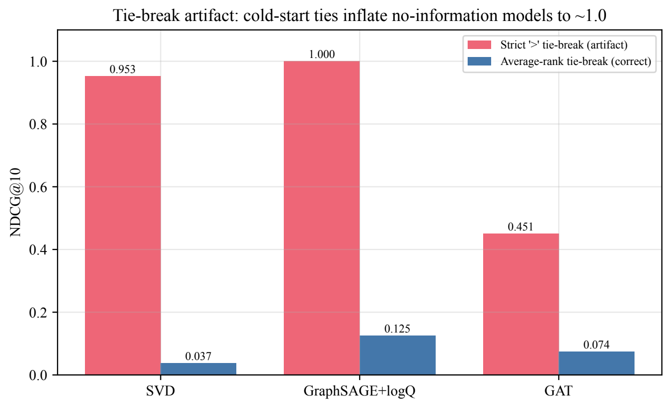

Figure 7: The tie-break artifact. Under a strict greater-than rule, degenerate cold-start models are placed first on tied candidates and inflate toward perfect scores; average-rank tie-breaking returns them to their true levels.

Table 6: Content company-feature ablation (temporal split, NDCG@10 with AUC in parentheses). Replacing the fixed-random company feature with the mean SBERT of each company's training-received patents does not help the learned models. Rows whose scores do not depend on the company feature (MostPop, NGCF, SVD, ...) are unchanged and omitted.

Model	Fixed-random feature	Content (mean-SBERT) feature
Text MLP	0.150 (0.574)	0.032 (0.399)
GraphSAGE	0.084 (0.473)	0.062 (0.481)
GAT	0.074 (0.519)	0.051 (0.504)
MostPop (reference)	0.197 (0.583)	

0.200–0.210, MostPop-IPC 0.219–0.228), while every genuinely learned model stays below — the strongest, a text MLP, reaches 0.148–0.163, and the graph models remain far lower (GraphSAGE 0.076–0.113, GAT 0.050–0.066, SVD 0.042–0.060). The collapse of learned models below the popularity baselines is therefore not specific to one split.

5.10. Architecture and hyperparameter sensitivity

A further objection is that the GNNs fail only because they are undersized or under-tuned. We therefore retrain GraphSAGE and GAT over a grid of embedding dimension (hidden/output size in {16/8, 32/16, 64/32, 128/64}) and learning rate ($\{0.005, 0.01, 0.02\}$) — 24 configurations in total (Table 8).

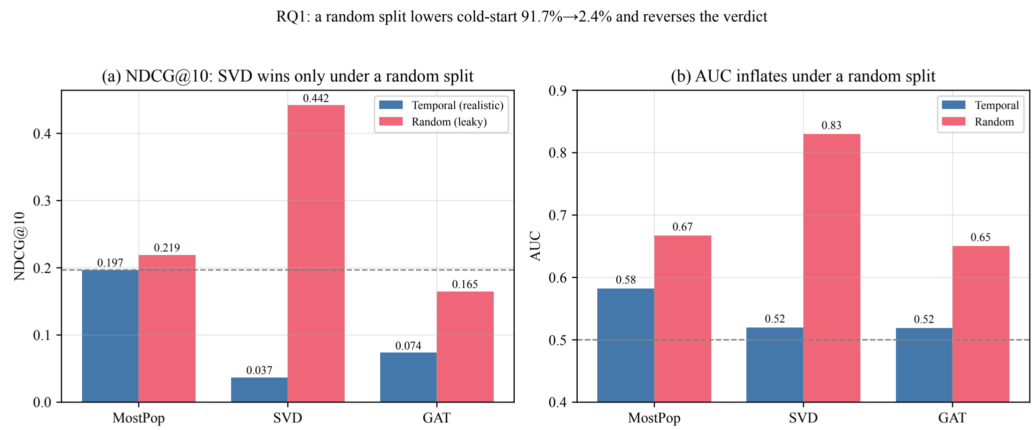

Figure 8: RQ1: temporal vs. random split. A random split lowers the cold-start rate (91.7%→2.4%) and reverses the verdict — (a) SVD becomes the best model by NDCG@10, and (b) AUC inflates across the board.

The best configuration reaches NDCG@10 $= 0.150$ (GraphSAGE), and GAT peaks at 0.110; both remain far below the MostPop baseline of 0.197, and most configurations are lower still. No reasonable architecture or learning-rate setting lifts a GNN to the popularity skyline, so the diagnosis is not an artifact of an under-capacity or under-tuned model.

6. Mitigations (RQ3)

We ask whether standard popularity-bias mitigations can lift a learned model above the popularity baselines, applying each symmetrically to both GNN backbones: popularity-debiased negative sampling, IPS / log-popularity re-ranking, the logQ correction, DropEdge, a sinusoidal transfer-edge time encoding, and the stacked combinations Debias+IPS and Time+IPS. Concretely, popularity-debiased sampling draws hard negatives with probability $P(c) \propto \mathrm{pop}_c^{\alpha}$ (exponent α; α=0 recovers uniform same-IPC sampling); IPS / log-popularity re-ranking penalizes the inference score as $\tilde{s}(p,c) = s(p,c) - \beta \log(\mathrm{pop}_c + 1)$ (penalty β); and the logQ correction subtracts the log negative-sampling probability from each candidate logit during training, $s(p,c) \leftarrow s(p,c) - \log Q(c)$, debiasing the sampled-softmax estimate. The gains are partial and unstable (Figure 9).

- On GAT, two mitigations produce a small but statistically significant and "honest" (above-chance-AUC) gain: the logQ correction (NDCG@10

Table 7: Rolling-origin robustness: NDCG@10 across three forward-chaining temporal folds (three seeds each). No learned model surpasses the popularity baselines at any cutoff.

Model	Fold 1 (test 70–80%)	Fold 2 (80–90%)	Fold 3 (90–100%)
MostPop-IPC	0.219	0.228	0.226
MostPop	0.200	0.209	0.210
NGCF	0.199	0.207	0.210
MLP (text)	0.158	0.148	0.163
GraphSAGE	0.113	0.092	0.076
GAT	0.066	0.057	0.050
SVD	0.047	0.060	0.042

Table 8: GNN architecture / hyperparameter sensitivity (temporal, NDCG@10, seed 0). GraphSAGE and GAT retrained over embedding dimension (hidden/output) and learning rate. The best of 24 configurations (0.150) remains below the MostPop baseline (0.197).

Backbone	hidden/out	$\eta = 0.005$	$\eta = 0.01$	$\eta = 0.02$
GraphSAGE	16/8	0.038	0.050	0.149
GraphSAGE	32/16	0.109	0.070	0.037
GraphSAGE	64/32	0.042	0.047	0.150
GraphSAGE	128/64	0.048	0.055	0.047
GAT	16/8	0.074	0.084	0.086
GAT	32/16	0.077	0.070	0.086
GAT	64/32	0.080	0.094	0.108
GAT	128/64	0.079	0.106	0.110

0.074→0.107, AUC 0.548) and the time encoding (0.074→0.085, AUC 0.521), both at Wilcoxon signed-rank Holm-adjusted $p = 0.047$ — yet both remain far below MostPop (0.197). GraphSAGE+logQ reaches 0.125 but is *not* significant after correction (high cross-seed variance, Holm-adjusted $p \approx 1.0$).

- IPS re-ranking raises NDCG@10 the most — GAT+IPS and GraphSAGE+Debias+IPS reach ≈0.157–0.158 — but at the cost of driving AUC to 0.417, below chance: the penalty demotes popular candidates wholesale, helping the metric when the positive is unpopular and hurting global discrimination. The β-sweep shows NDCG@10 rising mono-

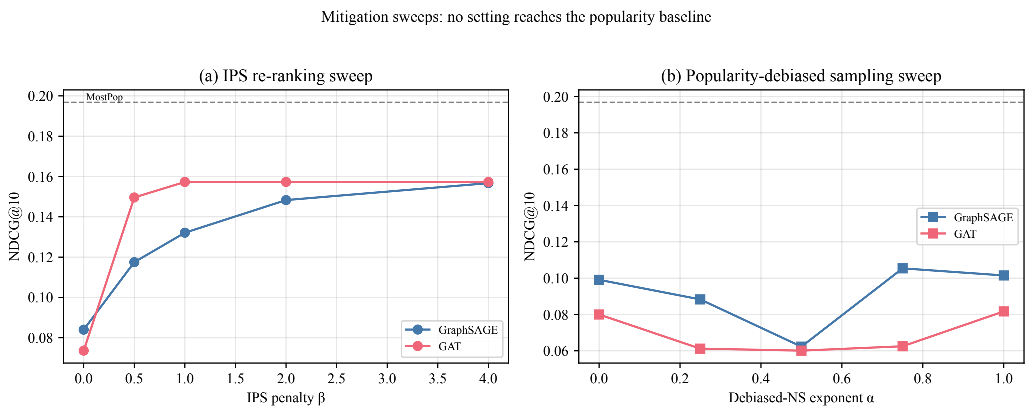

Figure 9: Mitigation sweeps. (a) IPS penalty β and (b) popularity-debiased sampling exponent α, per backbone. The dashed line is the MostPop skyline; no setting reaches it, and IPS gains coincide with below-chance AUC.

tonically with the penalty (GAT $0.074 \rightarrow 0.157$) exactly as AUC falls, confirming the trade-off is mechanical rather than a relevance gain.

- Popularity-debiased sampling (α) is noisy and non-monotone, never reaching the baseline. DropEdge produces small, non-significant changes; the time encoding is significant on GAT (above) but not on GraphSAGE.

No configuration in the grid reaches the popularity skyline; the mitigations relocate error rather than remove it.

7. Limitations and Threats to Validity

Construct validity.. Our supervision is realized transfer, an operationalization of latent demand; a model that perfectly predicted realized transfers would recover the historical (and biased) matching process. We therefore read the popularity-bias diagnosis partly as a property of the supervision signal.

Internal validity.. Transfer supervision is split strictly by registration date and message-passed only on the training split; popularity/recency/time features are train-only. Tie-breaking is average-rank, which we show is necessary. A residual threat is that citation edges are not time-filtered, which we expect to be minor relative to the transfer-edge leakage we control.

External validity.. All experiments use the Korean KIPRIS corpus. The popularity-bias and cold-start regime are properties of this market's transfer record; we cannot claim the same collapse would reproduce on a market with a different transfer density or company-size distribution. The diagnosis is established for one jurisdiction and is a hypothesis elsewhere.

Statistical-conclusion validity.. Ten seeds with Wilcoxon signed-rank + Holm–Bonferroni and bootstrap CIs over queries; we report effect sizes alongside adjusted p-values so a significant but negligible gain (e.g. logQ's +0.033 NDCG@10 on GAT, still below MostPop) is not over-read. The random-ranking floors are NDCG@10 ≈ 0.045 and AUC $= 0.5$; AUC near 0.5 with above-floor NDCG is not a contradiction but the signature of a popularity prior.

Temporal cutoff.. Our main results use a single 70/15/15 temporal split. We verify that the diagnosis is not specific to this cutoff with a three-fold rolling-origin evaluation (§5.9), in which no learned model beats the popularity baselines at any cutoff, and with the controlled random-split contrast (§5.7), which confirms that the realistic protocol is what exposes the failure. A finer-grained or longer rolling horizon remains possible future work.

Future work.. Replication on USPTO/EPO records and public link-prediction benchmarks; rolling-origin evaluation; content-aware company representations and cold-start-specific architectures; and treatment of the realized-transfer/latent-demand gap via exposure modeling.

Generalizability.. The quantitative verdict — the magnitude of the cold-start rate, the size of the popularity-baseline gap, and the temporal-vs-random reversal — is specific to the KIPRIS transfer record and its company-size distribution, and we do not claim the same ordering reproduces on markets with different transfer density. What we expect to generalize is methodological: the protocol (temporal split, same-class hard negatives, cold-start accounting, average-rank tie-break), the bias and cold-start diagnostics, and the strict-tie-break artifact are dataset-agnostic and apply to any sampled-ranking evaluation of cold-start-heavy link prediction.

8. Conclusion

We examined patent technology-transfer demand prediction as link prediction over a heterogeneous patent–citation–transfer graph, and asked whether learned models recover useful demand signal under a realistic protocol. Across heterogeneous GNNs (GraphSAGE, GAT), CF recommenders (LightGCN, NGCF), matrix factorization (SVD), structural heuristics (Common Neighbors, Adamic–Adar), and a text MLP, no learned model surpasses a simple popularity baseline when the evaluation uses a temporal split, same-IPC hard negatives, and an average-rank tie-break. The strongest configurations are not learned: MostPop-IPC (0.199), MostPop (0.197), and a popularity-equivalent NGCF (0.196); the strongest learned model, a text MLP, reaches only 0.150, and every graph-based model is lower (GraphSAGE+logQ 0.125, GAT+logQ 0.107), with AUC near chance (0.42–0.60). Two compounding causes account for this: popularity bias (98.8% of GraphSAGE and 57.8% of GAT errors are losses to a more-popular hard negative; ρ up to 1.0) and extreme patent cold-start (91.7% unseen; SVD 0.000 on new patents). The mitigations we evaluate yield only partial, unstable gains: logQ significantly improves GAT (0.074 $\rightarrow$ 0.107) yet stays below MostPop, and IPS re-ranking raises NDCG@10 to $\approx$0.157 only by driving AUC to 0.417. We additionally document an evaluation artifact — a strict tie-break inflates degenerate cold-start models to near-perfect scores ($\approx$0.95–1.00), corrected to 0.037 and 0.125 by average-rank — and show the diagnosis survives an unsampled full-pool metric. Finally, the protocol is decisive: under a random split the cold-start rate falls to 2.4% and matrix factorization becomes the single best method (0.442, AUC 0.830), reversing the temporal verdict. Our aim has been not a winning model but an evaluation protocol and a diagnosis — with reusable bias-diagnostic tools — to discipline future work on patent technology-transfer prediction and on cold-start-heavy link prediction more broadly.

Implications for research and practice.. For *research*, the results argue for reporting patent-recommendation and technology-transfer results under temporal splits with same-class hard negatives and explicit cold-start accounting, and for treating a popularity baseline and an AUC sanity check as mandatory reference points; near-perfect random-split numbers should be read as upper bounds, not as deployment estimates. For *practice*, technology-transfer intermediaries should not expect a learned ranker to outperform a simple popularity or recency heuristic on genuinely new patents under this market's data, and a popularity prior is a defensible, low-cost operational default; closing the gap will require content-aware company representations and exposure-aware supervision rather than a change of graph architecture alone.

Data availability

The patent data were obtained from the Korea Intellectual Property Rights Information Service (KIPRIS) through its paid API (KIPRIS Plus, `https://plus.kipris.or.kr`) and were subsequently cleaned and processed by the authors. Because redistribution of these data is restricted by the KIPRIS terms of use, the processed dataset is not deposited in a public repository; it is instead available from the authors by email upon reasonable request, subject to those terms. The preprocessing and evaluation code, the experiment scripts, and a script to regenerate the frozen SBERT embeddings from patent text are released, so that all tables and figures can be reproduced once the KIPRIS exports are obtained.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to improve the language, grammar, and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The figures are data visualizations generated by the authors' own analysis code directly from the experimental results, without generative AI.

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